Advantages

1. It has excellent frequency response. The gain is constant over the audio frequency range, which is the region of most importance for speech, music etc.,
2. Low Cost
3. The circuit is very compact

Disadvantages

1. The RC coupled amplifiers have low voltage and power gain
2. They have the tendency to become noisy with age, particularly in moist climates
3. Impedance matching is poor

Application

They are widely used as voltage amplifier, (e.g.) in the initial stages of public address system.

5.10.2 Direct-Coupled Amplifier

There are many applications in which extremely low frequency (less than 4 Hz) signals have to be amplified, (e.g.,) amplifying photoelectric current, thermo couple current, etc. The coupling devices such as capacitor and transformers cannot be used because the electrical sizes of these components become very large at extremely low frequencies. Under such situations, the output of one stage is directly connected to the next stage without any coupling device. This type of coupling is known as direct coupling.

Circuit details: Figure 5.24 shows the circuit of three stage direct coupled amplifier. It uses complementary transistors. Thus, the first stage uses npn transistor, the second stage uses pnp transistors and so on. This arrangement makes the design very simple. The output from the collector of first transistor T_1 is fed to the input of the second transistor T_2 and so on.

Operation: The weak signal is applied to the input of first transistor T_1. Due to transistor action, an amplified output is obtained across the collector load R_C of the transistor T_1.This voltage drives the base of the second transistor T_2 and amplified output is obtained across its collector load. In this way, direct coupled amplifier raises the strength of weak signal.

Advantages

1. The circuit arrangement is simple.
2. Low cost because of the absence of expensive coupling devices.

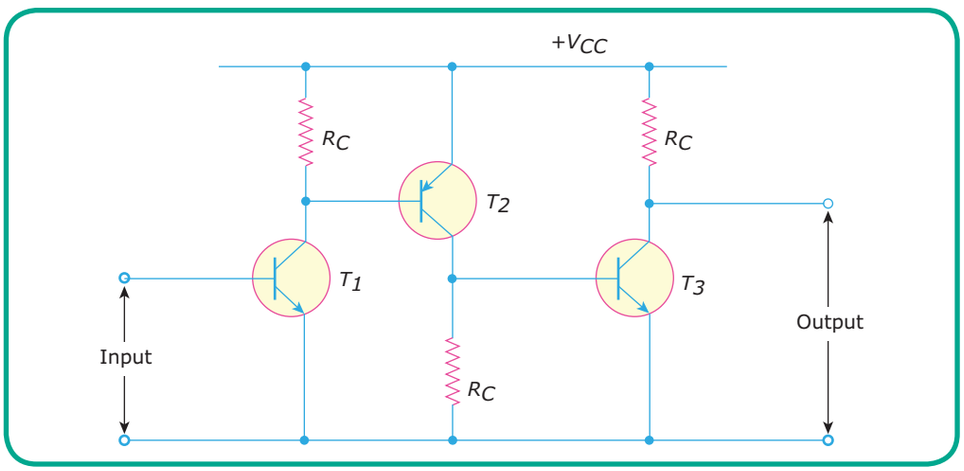


Figure 5.24 Stages of Direct Coupled Amplifier

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